

Basics 2: Vectors, Fields, and Coordinate Systems

I - Mathematical and computational foundations

Geometry and notation for Cartesian, cylindrical, toroidal, and field-aligned descriptions

Textbook draft, version 1.0. Plasma models are field theories. Correctly representing them requires a careful distinction among coordinates, basis vectors, vector components, metrics, and physical vectors. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Distinguish scalars, vectors, tensors, coordinates, and components.
- Use dot, cross, and triple products physically and algebraically.
- Represent vectors in Cartesian and cylindrical coordinates.
- Explain why curvilinear basis vectors vary with position.
- Use scale factors and Jacobians to compute distances, areas, and volumes.
- Interpret radial, poloidal, toroidal, parallel, and perpendicular directions.

2 Prerequisites and reading strategy

- Basics 1 or equivalent algebra and elementary trigonometry.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Scalars, vectors, tensors, and fields

A scalar such as density $\rho(\mathbf{x}, t)$ assigns one number to each point. A vector such as velocity $\mathbf{u}(\mathbf{x}, t)$ assigns a magnitude and direction. A tensor such as the pressure tensor $\mathbf{P}$ describes a direction-dependent linear relation.

The physical vector is independent of coordinates, but its components are not:

$$\mathbf{B} = B^1 \mathbf{e}_1 + B^2 \mathbf{e}_2 + B^3 \mathbf{e}_3. \quad (3.1)$$

Changing the basis changes the components while leaving the vector unchanged. This is why a data file must identify both component ordering and coordinate system.

4 Dot products, cross products, and projections

The dot product

$$\mathbf{a} \cdot \mathbf{b} = ab \cos \theta \quad (4.1)$$

measures projection and defines orthogonality. The component of $\mathbf{a}$ along a unit vector $\mathbf{b}$ is $a_{\parallel} = \mathbf{a} \cdot \mathbf{b}$.

The cross product

$$\mathbf{a} \times \mathbf{b} = ab \sin \theta \mathbf{n} \quad (4.2)$$

is normal to the plane of the two vectors. The Lorentz magnetic force is perpendicular to both velocity and magnetic field. The scalar triple product

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) \quad (4.3)$$

is an oriented volume and appears in coordinate Jacobians.

5 Cartesian coordinates

Cartesian coordinates use fixed orthonormal basis vectors:

$$\mathbf{r} = x\mathbf{e}_x + y\mathbf{e}_y + z\mathbf{e}_z, \quad d\ell^2 = dx^2 + dy^2 + dz^2. \quad (5.1)$$

Because the basis vectors do not vary with position, differentiation is comparatively simple. Cartesian coordinates are ideal for local wave derivations and code-verification cases, though they do not conform naturally to a torus.

6 Cylindrical coordinates and rotating basis vectors

Cylindrical coordinates (R, ϕ, Z) satisfy

$$x = R \cos \phi, \quad y = R \sin \phi, \quad z = Z. \quad (6.1)$$

The basis vectors are

$$\mathbf{e}_R = (\cos \phi, \sin \phi, 0), \quad (6.2)$$

$$\mathbf{e}_\phi = (-\sin \phi, \cos \phi, 0), \quad (6.3)$$

$$\mathbf{e}_Z = (0, 0, 1). \quad (6.4)$$

They obey

$$\frac{\partial \mathbf{e}_R}{\partial \phi} = \mathbf{e}_\phi, \quad \frac{\partial \mathbf{e}_\phi}{\partial \phi} = -\mathbf{e}_R. \quad (6.5)$$

The line element is

$$d\ell^2 = dR^2 + R^2 d\phi^2 + dZ^2, \quad (6.6)$$

so an angular differential represents physical distance $R d\phi$.

7 General coordinates, metric factors, and Jacobians

For orthogonal coordinates (q^1, q^2, q^3) with scale factors h_i ,

$$d\ell^2 = \sum_i h_i^2 (dq^i)^2, \quad dV = h_1 h_2 h_3 dq^1 dq^2 dq^3. \quad (7.1)$$

In cylindrical coordinates $(h_R, h_\phi, h_Z) = (1, R, 1)$, so $dV = R dR d\phi dZ$. Omitting the Jacobian factor changes integrals, conservation laws, and matrix symmetry.

In nonorthogonal coordinates, the metric tensor

$$g_{ij} = \frac{\partial \mathbf{r}}{\partial q^i} \cdot \frac{\partial \mathbf{r}}{\partial q^j} \quad (7.2)$$

determines distances and inner products. Covariant and contravariant components are then distinct.

8 Toroidal and magnetic coordinates

A simple circular torus may be written

$$R = R_0 + r \cos \theta, \quad (8.1)$$

$$Z = r \sin \theta, \quad (8.2)$$

where θ is poloidal and ϕ toroidal. Real stellarator surfaces require Fourier representations in both angles.

Magnetic coordinates commonly use a surface label ψ together with angles. A magnetic surface satisfies

$$\mathbf{B} \cdot \nabla \psi = 0. \quad (8.3)$$

The field-aligned unit vector is $\mathbf{b} = \mathbf{B}/B$, and any vector decomposes as

$$\mathbf{a} = (\mathbf{a} \cdot \mathbf{b})\mathbf{b} + [\mathbf{a} - (\mathbf{a} \cdot \mathbf{b})\mathbf{b}]. \quad (8.4)$$

The terms parallel and perpendicular always refer to a specified local magnetic field.

A component called “radial” may mean cylindrical R , minor radius r , or flux coordinate ψ . Never infer the meaning from the word alone.

9 Computational laboratory

Generate points on a circular torus, convert between (r, θ, ϕ) and Cartesian coordinates, and verify basis orthogonality. Approximate the torus volume using the correct Jacobian and compare with $2\pi^2 R_0 a^2$. Then omit the Jacobian deliberately and quantify the error. Implement identical coordinate conventions in Python and MATLAB.

10 Summary

- Vectors are coordinate independent, but their components and bases are not.
- Curvilinear basis vectors and metric factors vary with position.
- Jacobians are essential to physical integration and numerical conservation.
- Toroidal, poloidal, radial, parallel, and perpendicular directions are distinct.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
$\mathbf{e}_i$	basis vector	Direction associated with coordinate q^i .
h_i	scale factor	Converts coordinate differential to physical length.
g_{ij}	metric tensor	Inner products of coordinate basis vectors.
R, ϕ, Z	cylindrical coordinates	Radius, azimuth, and vertical coordinate.
$\mathbf{b}$	field unit vector	$\mathbf{B}/B$.

11 Exercises

1. Prove the cylindrical-basis derivative identities.
2. Compute the physical length associated with $d\phi$ at two different major radii.

3. Decompose a given vector into parallel and perpendicular components.
4. Derive the volume of a circular torus from a coordinate mapping.
5. Explain the difference between a coordinate basis and an orthonormal physical basis.

12 References

References

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